

Natalie M. Allen

Department of Forestry and Natural Resources, Purdue University
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EDUCATION

Purdue University, West Lafayette, Indiana USA January 2023 — Present
Ph.D. Candidate in Evolutionary Biology and Conservation Genomics
Marquette University, Milwaukee, Wisconsin USA August 2019 — December 2022
B.S. in Biological Sciences with Comprehensive Honors, *magna cum laude*
Minor in Environmental Studies

RESEARCH EXPERIENCE

Department of Forestry and Natural Resources, Purdue University West Lafayette, Indiana USA
Ph.D Candidate, DeWoody Lab January 2023 - Present

- Conducted the first population genomic study of the western chicken turtle (*Deirochelys reticularia miaria*), which revealed unexpected population structure that may inform the establishment of conservation units.
- Led a study on gray whale (*Eschrichtius robustus*) evolutionary history and contemporary population dynamics using low coverage whole genome resequencing data.
- Investigated drivers of evolutionary divergence across vertebrates using whole-genome comparison of sister taxa.
- Used vertebrate pangenome assemblies to assess structural variants in Montezuma Quail and related species.
- Tested a short-read pangenome pipeline for use in conservation genetics.
- Assembled a novel vertebrate reference genome.
- Developed and implemented bioinformatic pipelines for genomic data analysis, including comparative genomics, demographic inference, and population genetics.

Department of Biological Sciences, Marquette University Milwaukee, Wisconsin USA
Undergraduate Researcher, Gamble Evolutionary Biology Lab September 2019 – December 2022

- Led an independent project to study the evolution of visual opsin genes across geckos using phylogenetic trees, selection tests, and molecular clock analysis (August 2020 – December 2022).
- Completed DNA extraction, PCR, gel electrophoresis, multi-sequence alignment, and phylogenetic analysis to identify hybrid geckos (September 2019 – March 2020).

Department of Environmental Engineering Sciences, University of Florida Gainesville, Florida USA
High School Research Assistant, Angelini Ecology Lab August 2017 – May 2019

- Researched biomagnification of polychlorinated biphenyls through trophic levels by counting Red Drum pharyngeal teeth, examining fish stomach contents, and cataloguing sources for a meta-analysis paper.

Department of Wildlife Ecology and Conservation, University of Florida Gainesville, Florida USA
High School Research Assistant, Austin Conservation Genetics Lab August 2015 – August 2017

- Conducted DNA extraction, PCR, and gel electrophoresis to support avian malaria research.

SKILLS

- **Bioinformatics:** Pre-processing, variant calling, and population genomic analysis of whole genome resequencing data; genome assembly; phylogenomics and comparative genomics; pangenomics using PGGB and Minigraph-Cactus; Nextflow pipeline construction; R; Python; shell scripting; ArcGIS Pro; R and Python for GIS.
- **Laboratory Genetics:** HMW DNA extraction, cDNA synthesis, PCR, gel electrophoresis, DNA cloning, RT-qPCR.

- **Field Work:** Field identification of North American vertebrates (especially herpetofauna), freshwater turtle trapping and tissue sampling.

SCIENTIFIC PUBLICATIONS

Natalie M. Allen, Andrew J. Mularo, Jong Yoon Jeon, Alvaro G. Hernandez, Ethan S. Streckfuss, Wade A. Ryberg, Toby J. Hibbitts, Brandon C. Bowers, Paul Crump, and J. Andrew DeWoody. (2025) Whole genome resequencing data provide high-resolution conservation insights into population structure and life history. *Conservation Genetics*. *In review*.

Jong Yoon Jeon, Andrew M. Mularo, **Natalie M. Allen**, Erangi Heenkenda, John W. Bickham, Andrew N. Black, J. Andrew DeWoody. (2025) Genomic metrics can inform formal listing processes associated with the conservation of threatened and endangered species. *Trends in Ecology and Evolution*. *In review*.

Jong Yoon Jeon, **Natalie M. Allen**, Andrew N. Black, J. Andrew DeWoody. (2025) Short-Read Pangenomes and Their Potential Utility in Population and Conservation Genomics. *Molecular Ecology Resources* e70060. <https://doi.org/10.1111/1755-0998.70060>

Andrew N. Black, Jong Yoon Jeon, Andrew M. Mularo, **Natalie M. Allen**, Erangi Heenkenda, Julia C. Buchanan-Schwanke, John W. Bickham, Zachary E. Lowe, J. Andrew DeWoody. (2024) Thematic layers of genomic susceptibility for conservation assessments: introgressive hybridization as an example. *Molecular Ecology* e17582. <https://doi.org/10.1111/mec.17582>

CONFERENCE AND SEMINAR TALKS

- Whole genome resequencing data provide high-resolution conservation insights into population structure and life history. *Evolution* 2025, Athens, Georgia, USA.
- Conservation insights from population genomics in an elusive turtle. Department of Biological Sciences *Ecolunch* 2025, Purdue University
- Population genomics of the Pacific gray whale. *Evolution* 2024, Montreal, Canada.
- Population genomics of the gray whale. *Midwest Population Genetics* 2023, University of Michigan.

POSTER PRESENTATIONS

- Population genomics can inform conservation efforts in an elusive turtle. *Midwest Population Genetics* 2024, Indiana University.
- Conservation genomics of the western chicken turtle. Department of Forestry and Natural Resources 2023 poster symposium, Purdue University.
- The evolution of gecko vision. Department of Biological Sciences 2022 Summer Research Symposium, Marquette University.
- The evolution of gecko vision. *Evolution* 2022, Cleveland, Ohio, USA.
- Evidence of molecular convergence and positive selection on visual opsin proteins in geckos. Department of Biological Sciences 2022 Undergraduate Research Day, Marquette University.
- Testing for molecular convergence and selection on gecko opsin genes. Honors Undergraduate Research Fair 2021, Marquette University.
- The evolution of gecko vision: Molecular convergence in visual opsin proteins. Department of Biological Sciences 2021 Undergraduate Research Day, Marquette University.

MONETARY AWARDS

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| • Welder Wildlife Foundation Research Fellowship, 2026 | \$26,400 |
| • D. Woods Thomas Memorial International Support Fund, 2025 | \$1500 |
| • Purdue Institute for a Sustainable Future Travel Award, 2024 | \$650 |

- Marquette Biological Sciences Conference Travel Award, 2022 \$1500
- Marquette Biological Sciences Summer Research Program, 2022 \$4000
- Marquette Honors Summer Research Fellowship, 2021 \$3500

WORK AND TEACHING

Purdue University

Graduate Research Fellow, Welder Wildlife Foundation January 2026 - Present
Graduate Teaching Assistant, Statistics for Natural Resources January 2026 - Present
Graduate Research Assistant, DeWoody Lab January 2023 - December 2025

Marquette University

Teaching Assistant, Principals of Biological Investigation Fall 2022
Teaching Assistant, Evolutionary Biology Spring 2022
Teaching Assistant, Cellular Biology Fall 2021
Animal Caretaker, Geckos in Gamble Lab Aug. 2020 - Dec. 2022
 • *Manager* (May 2022 - Dec. 2022)

Varsity Tutors

Tutor, ACT, SAT, Algebra, Precalculus, Calculus 1 July 2019 - Dec. 2022

Independent Tutoring

Tutor, English for ESL students, Algebra Feb. 2016 - Dec. 2022

UNDERGRADUATE MENTEES

- Ethan Streckfuss August 2024 - May 2025
 - Poster presentation: Genomic insights into the conservation of the western chicken turtle, Spring Undergraduate Research Expo 2025, Purdue University, USA
- Brady Kahner (co-mentored) August 2024 - Present
- Megan Johnson January 2023 - May 2024
 - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Fall Undergraduate Research Expo 2023, Purdue University, USA
 - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Society for Molecular Biology and Evolution "Molecular Evolution in Small Populations" 2023, Princeton University, USA
- Rafaela Besse January 2023 - December 2023
 - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Fall Undergraduate Research Expo 2023, Purdue University, USA
 - Poster presentation: Nuclear copies of Mitochondrial DNA fragments (NUMTs) in the Gray Whale Genome, Society for Molecular Biology and Evolution "Molecular Evolution in Small Populations" 2023, Princeton University, USA

ATTENDED TRAININGS

- Workshop: Communicating Science to Diverse Audiences, Evolution Meeting June 2025
- Workshop: GIS and Data Science, Purdue University October 2024
- Workshop: Designing Nextflow pipelines, Purdue University January - May 2024
- Course: Analysis of high throughput data for population genomics, University of Copenhagen (virtual) August 2023
- Workshop: SLiM forward evolutionary simulation software, Albuquerque, New Mexico June 2023
- Training: RT-qPCR, Purdue FNR Genetics Lab May 2023

ARTICLES

- **Allen, N.** (2020, July 30). *Allen: MU must offer remote learning, teaching options for Fall Semester*. Marquette Wire. <https://marquettewire.org/4034276/featured/allen-mu-must-offer-remote-learning-teaching-options-for-fall-semester/>

PEER REVIEW CONTRIBUTIONS

- Science (co-reviewer)
- Molecular Ecology
- Molecular Ecology Resources

SERVICE AND LEADERSHIP

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| • <i>College of Agriculture Representative</i> , FNR Graduate Student Council | May 2025 - Present |
| • <i>Judge</i> , Fall Undergraduate Research Expo | 2024 |
| • <i>Organizer</i> , Lab Bioinformatics Coffee Hours | Jan. 2024 - Present |
| • <i>Volunteer</i> , Wisconsin Humane Society | 2022 |
| • <i>Founder</i> , Biology Research Achievement Mentorship Program (RAMP) | 2022 |
| • <i>Interviewer</i> , Building DEI in Marquette Honors Admissions | 2022 |
| • <i>Vice President</i> , Marquette Waterski Team | 2022 |
| • <i>Women's Captain</i> , Marquette Waterski Team | 2021 |
| • <i>Volunteer</i> , River Revitalization Foundation | 2020 - 2021 |
| • <i>Volunteer</i> , Milwaukee Public Museum | 2019 - 2020 |